Supplement to Blockwise Occupancy Inference under Weak Dependence with Record-Based Persistence Thresholds

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This supplement contains the coupling and CHARN proofs and the additional numerical comparisons. References to unprefixed sections, equations, and results refer to the main manuscript.

S1 Vector coupling lemma and classification proof

Lemma S1.1 (Vector Berbee coupling)**.** *Let $Y_1, \dots, Y_m$ be random elements of standard Borel spaces. On an extension of the probability space there exist mutually independent $Y_1^*, \dots, Y_m^*$, with $\mathcal{L}(Y_j^*) = \mathcal{L}(Y_j)$ for every j, such that*

$$\mathbb{P}\{(Y_1, \dots, Y_m) \neq (Y_1^*, \dots, Y_m^*)\} \leq \sum_{j=2}^{m} \beta\{\sigma(Y_1, \dots, Y_{j-1}), \sigma(Y_j)\}.$$

Proof. We compare the joint vector law with the product of its marginals. Write $\mu_j = \mathcal{L}(Y_1, \dots, Y_j)$, $\nu_j = \mathcal{L}(Y_j)$, and $\pi_j = \bigotimes_{i=1}^{j} \nu_i$. With the total-variation convention used in this paper,

$$d_{\mathrm{TV}}(\mu_j, \mu_{j-1} \otimes \nu_j) = \beta\{\sigma(Y_1, \dots, Y_{j-1}), \sigma(Y_j)\}.$$

Tensoring both measures with the same probability law does not change total variation. We can therefore use the triangle inequality to obtain the recursion

$$d_{\mathrm{TV}}(\mu_j, \pi_j) \leq d_{\mathrm{TV}}(\mu_{j-1}, \pi_{j-1}) + \beta\{\sigma(Y_1, \dots, Y_{j-1}), \sigma(Y_j)\}.$$

Iterating this inequality, we bound $d_{\mathrm{TV}}(\mu_m, \pi_m)$ by the displayed sum.

It remains to realize this comparison as a coupling. A maximal coupling of the two vector laws μ_m and π_m exists on standard Borel spaces and has mismatch probability $d_{\mathrm{TV}}(\mu_m, \pi_m)$. The second vector has product law π_m, so its coordinates are mutually independent and have the required marginals. This is the vector form of Berbee's coupling lemma (Berbee, 1979). □

Proof of Theorem 3.8. We work first with the independent Bernoulli surrogate $S^* = \sum_{j=1}^{m} Y_j^*$

from Theorem 3.2. For $\lambda > 0$,

$$\mathbb{E}e^{\lambda S^*} = \prod_{j=1}^{m} \{1 - p_{b,j} + p_{b,j} e^{\lambda}\}.$$

To replace the individual success probabilities by their average, we use the concavity of the logarithm of each factor as a function of $p_{b,j}$. Jensen's inequality gives

$$\mathbb{E}e^{\lambda S^*} \le \{1 - \bar{p}_b + \bar{p}_b e^{\lambda}\}^m.$$

Optimizing the resulting Chernoff bound gives

$$\mathbb{P}(S^* \ge mx) \le e^{-mD_{\mathrm{KL}}(x\|\bar{p}_b)} \quad (x > \bar{p}_b), \qquad \mathbb{P}(S^* \le mx) \le e^{-mD_{\mathrm{KL}}(x\|\bar{p}_b)} \quad (x < \bar{p}_b),$$

at the integer values used below.

We now identify the error event for each label. For the reference label, the error event is $\{S^* \ge a_m{+}1\} = \{S^* \ge mu_m\}$. For the elevated label, it is the union of $\{S^* \le a_m\} = \{S^* \le m\ell_m\}$ and $\{S^* \ge k_m\} = \{S^* \ge mv_m\}$. For the high-occupancy label, it is $\{S^* \le k_m{-}1\} = \{S^* \le mw_m\}$. Applying the corresponding tail bounds gives the three expressions, with a union bound for the two elevated-label errors. The lower-tail bound is obtained by repeating the exponential argument with $\lambda < 0$. Finally, we return to the dependent count. Theorem 3.2 changes the probability of each complete error event by at most one $\varepsilon_{\mathrm{mix},b}$, which proves (31)-(33). □

S2 CHARN drift, minorization, and initialization

Proof of Proposition 7.4. We first construct a quadratic drift function adapted to the lag shifts. Write $u(z) = (z_1^2, \dots, z_p^2)^\top$ and choose $\rho(B_c) < \lambda_0 < 1$. The positive vector defined by the convergent Neumann series

$$w = (I - B_c^\top/\lambda_0)^{-1}\mathbf{1}_p = \sum_{k\ge 0} (B_c^\top/\lambda_0)^k \mathbf{1}_p \tag{S1}$$

satisfies $w \ge \mathbf{1}_p$ and $B_c^\top w = \lambda_0(w - \mathbf{1}_p) \le \lambda_0 w$. Using the centered unit-variance innovations and equation (59), we obtain

$$\mathbb{E}\{u(Z_t) \mid Z_{t-1} = z\} \le A\mathbf{e}_1 + B_c u(z).$$

Hence $W(z) = 1 + w^\top u(z)$ obeys

$$P_\vartheta W(z) \le 1 + Aw_1 + \lambda_0 w^\top u(z) = \lambda_0 W(z) + (1 + Aw_1 - \lambda_0)$$

uniformly in the parameter.

We next verify the small-set condition. Fix a sublevel set C_R and a target rectangle $K_L = [-L, L]^p$. Write $y_1, \dots, y_p$ for the next p generated observations and $z^{(k)} = (y_k, z_1^{(k-1)}, \dots, z_{p-1}^{(k-1)})$. The triangular change of variables from innovations to observations has Jacobian $\prod_{k=1}^{p} V_\eta(z^{(k-1)})$. The transition density, with terminal coordinates reversed, is consequently

$$\prod_{k=1}^{p} \frac{1}{V_\eta(z^{(k-1)})} f\left(\frac{y_k - T_\theta(z^{(k-1)})}{V_\eta(z^{(k-1)})}\right). \tag{S2}$$

When $z \in C_R$, the terminal state belongs to K_L, and the parameter ranges over its compact set, every intermediate state belongs to a common compact set. Joint continuity and the positive

lower scale bound give

$$0 < \underline{v} \le V_\eta(z^{(k)}) \le \overline{v} < \infty, \qquad |T_\theta(z^{(k)})| \le \overline{T}.$$

The innovation arguments therefore remain in a compact interval. The minimum of f on it is positive, so equation (S2) has a uniform positive lower bound on $C_R \times K_L$. Taking normalized Lebesgue measure on K_L, we obtain the p-step minorization, with a measure independent of the parameter.

For $p+1$ transitions, we integrate the first generated observation over a fixed compact interval of positive length. The same compactness argument applies to all $p+1$ factors and supplies a positive lower bound on the same terminal rectangle. This gives the consecutive minorization on a common target set. Finally, positivity of the p-step density on all of $\mathbb{R}^p$ proves Lebesgue irreducibility. In particular, choosing the target rectangle to contain a nonempty sublevel set also gives the required aperiodicity. □

Proof of Proposition 7.5. We choose $\zeta > 0$ sufficiently small that $\lambda_0 = (1+\zeta)c_T^2 + c_V < 1$. The growth bounds then give

$$\mathbb{E}(X_t^2 \mid X_{t-1} = x) = T_\theta(x)^2 + V_\eta(x)^2 \le (1+\zeta^{-1})A_T^2 + A_V + \lambda_0 x^2.$$

Thus $W(x) = 1 + x^2$ satisfies the stated drift inequality. For the one- and two-step compact minorizations, we specialize the compact-density argument in the preceding proof to $p = 1$. □

Proof of Proposition 7.6. The common drift and small-set bounds allow us to apply the drift-minorization theorem (Meyn and Tweedie, 2009, Chapters 15–16). We obtain uniform geometric ergodicity, with a unique invariant law π_ϑ satisfying

$$\|P_\vartheta^h(z,\cdot) - \pi_\vartheta\|_{\mathrm{TV}} \le MW(z)\bar{\rho}^h, \qquad \pi_\vartheta(W) \le b/(1-\lambda), \tag{S3}$$

with constants uniform over the parameter family. If the rate is initially obtained for a p-step skeleton, we write $h = pn + s$, $0 \le s < p$, and use contraction under P_ϑ^s. Then $\rho_p^n \le \rho_p^{-1}(\rho_p^{1/p})^h$, which gives the claimed raw-time rate after increasing M.

We now pass from transition probabilities to absolute regularity. For a stationary Markov chain on a standard Borel state space, the conditional-law representation and the Markov property give

$$\beta_Z(h) \le \int \|P_\vartheta^h(z,\cdot) - \pi_\vartheta\|_{\mathrm{TV}}\, \pi_\vartheta(\mathrm{d}z). \tag{S4}$$

Indeed, the future path is obtained from its initial state by a common probability kernel, and total variation contracts under that kernel. The bound therefore controls the full past and future sigma-fields, as required. We integrate equation (S3) to obtain $C = Mb/(1-\lambda)$. Since X_t is a coordinate of Z_t, $\beta_X(h) \le \beta_Z(h)$. Proposition 3.1 and Theorem 3.2 then give the diagnostic count bound with raw support r_{eff}. □

Proof of Proposition 7.8. At time $\tau + s$, the state law is μP_ϑ^s. We use the same conditional-law representation as in the stationary case, inserting π_ϑ in the triangle inequality. This gives

$$\begin{aligned}\beta_{\mu,\vartheta}(s,h) &\le \int \|P_\vartheta^h(z,\cdot) - \mu P_\vartheta^{s+h}\|_{\mathrm{TV}}\, \mu P_\vartheta^s(\mathrm{d}z) \\ &\le M\bar{\rho}^h\, \mu P_\vartheta^s(W) + M\mu(W)\bar{\rho}^{s+h}.\end{aligned}$$

The drift inequality implies $\mu P_\vartheta^s(W) \le \mu(W) + b/(1-\lambda)$. Therefore $C_\mu = M\{2\mu(W) + b/(1-\lambda)\}$ is a uniform finite constant in equation (65).

We turn next to the marginal exceedance probabilities. Let $K_{r_{\mathrm{eff}}}$ map the state immediately before a diagnostic's raw support to its exceedance indicator. For $d = t - r_{\mathrm{eff}} - \tau \geq 0$, contraction under this kernel gives

$$|\mathbb{P}_\mu(e_t = 1) - p_\vartheta| \leq \|\mu P_\vartheta^d - \pi_\vartheta\|_{\mathrm{TV}} \leq M\mu(W)\bar{\rho}^d.$$

This is the required marginal bound.

To compare the count with its stationary binomial law, we first use vector coupling. It gives the Poisson-binomial approximation with allowance $(m-1)C_\mu\bar{\rho}^{q-r_{\mathrm{eff}}+1}$. We then change the marginal parameters of the independent Bernoulli variables to p_ϑ at cost at most

$$\sum_{j=1}^{m} |\mathbb{P}_\mu(e_{t_{b,j}} = 1) - p_\vartheta| \leq M\mu(W)\bar{\rho}^{d_b}\frac{1-\bar{\rho}^{mq}}{1-\bar{\rho}^q}.$$

Combining these two comparisons by the triangle inequality proves equation (67). □

S3 Additional fitted-model comparisons

Table S1 gives the remaining competitors from the same fitted-reference experiment. The complete CSV includes replication-level Monte Carlo standard errors for each reported probability.

Table S1: Additional comparators under the same reference-block calibration. The definitions and error budgets are those of Section 8.

Scenario / method	$0 \to +$	$0 \to 2$	$1 \to 1$	$1 \to 2$	$2 \to 2$
Gaussian mean 0.40 / Capped energy	0.050	0.009	0.363	0.363	0.921
Gaussian mean 0.40 / Interval scan	0.048	0.010	0.000	1.000	0.996
Gaussian mean 0.40 / Run statistic	0.050	0.010	0.022	0.963	0.426
Gaussian mean 0.55 / Capped energy	0.050	0.010	0.359	0.360	1.000
Gaussian mean 0.55 / Interval scan	0.051	0.011	0.000	1.000	1.000
Gaussian mean 0.55 / Run statistic	0.053	0.010	0.033	0.953	0.832
Student mean 0.40 / Capped energy	0.051	0.011	0.353	0.348	0.974
Student mean 0.40 / Interval scan	0.051	0.012	0.002	0.996	0.999
Student mean 0.40 / Run statistic	0.050	0.009	0.057	0.917	0.552
Student mean 0.55 / Capped energy	0.051	0.010	0.359	0.332	1.000
Student mean 0.55 / Interval scan	0.051	0.009	0.004	0.995	1.000
Student mean 0.55 / Run statistic	0.048	0.009	0.067	0.902	0.914
Gaussian scale 1.50 / Capped energy	0.051	0.009	0.358	0.364	0.415
Gaussian scale 1.50 / Interval scan	0.053	0.010	0.000	1.000	0.210
Gaussian scale 1.50 / Run statistic	0.050	0.010	0.023	0.968	0.157
Student scale 1.50 / Capped energy	0.052	0.010	0.350	0.356	0.039
Student scale 1.50 / Interval scan	0.051	0.010	0.004	0.996	0.022
Student scale 1.50 / Run statistic	0.048	0.008	0.059	0.913	0.019
Gaussian alternating 0.55 / Capped energy	0.049	0.010	0.349	0.365	1.000
Gaussian alternating 0.55 / Interval scan	0.050	0.010	0.001	1.000	0.028
Gaussian alternating 0.55 / Run statistic	0.050	0.013	0.031	0.957	0.860

S4 Proof of the numerical mixing envelope

Proof of Proposition 7.9. Use the same innovation for chains started at x and y. Conditional on their current states, centering and unit variance give

$$\mathbb{E}\{|X_{n+1}^x - X_{n+1}^y|^2 \mid X_n^x, X_n^y\} = |T(X_n^x) - T(X_n^y)|^2 + |V(X_n^x) - V(X_n^y)|^2 \le \rho_c^2 |X_n^x - X_n^y|^2.$$

Iteration and Cauchy–Schwarz imply $\mathbb{E}|X_n^x - X_n^y| \le \rho_c^n |x - y|$.

For the location-scale density $f_{a,s}(z) = s^{-1} f((z-a)/s)$, differentiation and a change of variable give

$$\tfrac{1}{2} \int |\partial_a f_{a,s}(z)| \, \mathrm{d}z = A_f/s, \qquad \tfrac{1}{2} \int |\partial_s f_{a,s}(z)| \, \mathrm{d}z = B_f/s.$$

Interpolate first in location and then in scale, keeping every scale at least v_*. The fundamental theorem of calculus, followed by Tonelli's theorem, yields

$$d_{\mathrm{TV}}\{P(x,\cdot), P(y,\cdot)\} \le \{A_f |T(x) - T(y)| + B_f |V(x) - V(y)|\}/v_* \le L_P |x - y|.$$

Mixture convexity of total variation applied to the synchronously coupled states at time $h-1$ now gives

$$d_{\mathrm{TV}}\{P^h(x,\cdot), P^h(y,\cdot)\} \le L_P \rho_c^{h-1} |x - y|.$$

Invariance of π lets us integrate the second starting state against π. Integrating the first against π as well gives

$$\int d_{\mathrm{TV}}\{P^h(x,\cdot), \pi\} \, \pi(\mathrm{d}x) \le L_P \rho_c^{h-1} \iint |x - y| \, \pi(\mathrm{d}x) \pi(\mathrm{d}y) \le 2 L_P \sqrt{v_2} \rho_c^{h-1}.$$

The stationary Markov representation of absolute regularity bounds $\beta_X(h)$ by the left side. This proves (69). Integration of the first starting state against μ instead gives (70). A future path conditional on its first state is generated by a common Markov kernel, and total variation contracts under this kernel. The bounds therefore extend from the state to the whole subsequent path.

For the stated numerical constants, differentiating the reference mean gives $T'(x) = .45 - .25(1 - x^2)e^{-x^2/2}$, whose maximum absolute value is $.45 + .5e^{-3/2}$. Also $|V'(x)| \le \sqrt{.12}$. Since $|T(x)| \le .45|x|$, stationarity and the second-moment recursion give $\pi(x^2) \le .70 + (.45^2 + .12)\pi(x^2)$. The quadratic drift condition ensures this moment is finite, so rearranging yields v_2. For either symmetric unimodal density, $A_f = f(0)$. The function $xf(x)$ on the positive half-line increases up to $x = 1$ for the Gaussian density and $x = \sqrt{3/5}$ for the standardized Student density, then decreases to zero. Its total variation gives the displayed formulas for B_f. Substitution gives Table 1. □

S5 Window, count, and plug-in arguments

Proof of Proposition 3.1. We begin by identifying the raw-data sigma-fields on which the diagnostics depend. For every integer s,

$$\sigma(D_{u,r} : u \le s) \subseteq \mathcal{F}_{-\infty}^{s}, \qquad \sigma(D_{u,r} : u \ge s + h) \subseteq \mathcal{F}_{s+h-r+1}^{\infty}.$$

By monotonicity of $\beta(\mathcal{A}, \mathcal{B})$ in each sigma-field, these inclusions give $\beta_D(h) \le \beta_X(h - r + 1)$ when $h \ge r$. Since e_t is a measurable function of $D_{t,r}$, $\beta_e(h) \le \beta_D(h)$. For the sampled sequence, we replace the diagnostic lag h by the raw-time lag hq and obtain (12).

We use the same argument for the outer decisions, now taking account of the full block

support. All decisions with block index at most zero are measurable with respect to $\mathcal{F}_{-\infty}^{s_0+(m-1)q}$, whereas those with index at least h are measurable with respect to $\mathcal{F}_{s_0+h\nu-r+1}^{\infty}$. The separation between these boundary indices is

$$(s_0 + h\nu - r + 1) - \{s_0 + (m-1)q\} = h\nu - L + 1.$$

Applying sigma-field monotonicity to these two fields gives (13). □

Proof of Proposition 3.5. We first compare the joint law of $(e_{t_{b,i}}, e_{t_{b,j}})$ with the product of its marginals. Their total-variation distance is at most $\beta_X\{(j-i)q - r + 1\}$. Since $(u, v) \mapsto uv$ takes values in $[0, 1]$, the difference between its expectations under these two laws is bounded by the same total-variation distance, which proves (19).

To pass from individual covariances to the count variance, we use the identity

$$\mathbb{V}(E_b) - V_{\mathrm{ind},b} = 2 \sum_{1 \le i < j \le m} \mathrm{Cov}(e_{t_{b,i}}, e_{t_{b,j}}).$$

We group the pairs according to $h = j - i$ to obtain (20). Under geometric mixing,

$$\sum_{h=1}^{m-1} (m-h) C\rho^{hq-r+1} \le Cm\rho^{q-r+1} \sum_{h=0}^{\infty} \rho^{hq} = \frac{Cm\rho^{q-r+1}}{1-\rho^q},$$

so multiplication by two yields (21). □

Proof of Theorem 3.6. For the sums over the residue classes, let

$$S_a = \sum_{j=0}^{n_a-1} e_{a+jq}, \qquad a = 1, \dots, q.$$

Then $S_m = \sum_a S_a$ and $m = \sum_a n_a$. If $S_m - mp \ge x$, at least one class satisfies $S_a - n_a p \ge x/q$. A union bound therefore gives

$$\mathbb{P}(S_m - mp \ge x) \le \sum_{a=1}^{q} \mathbb{P}\{S_a \ge u_a(x)\}.$$

Within class a, successive diagnostics are q time points apart. Theorem 3.2 bounds its count law by $\mathrm{Bin}(n_a, p)$ with total-variation error at most $(n_a - 1)\beta_X(q - r + 1)$. We now sum the classwise errors. Using $\sum_a (n_a - 1) = m - q$, we obtain (23). For the lower tail, we repeat the argument with $S_m - mp \le -x \Rightarrow S_a - n_a p \le -x/q$ for at least one a. No independence between different residue classes is needed. □

Proof of Proposition 7.2. We write $h_n = \widehat{\vartheta} - \vartheta_0$. The estimation error enters the central sequence through a Taylor expansion. Under the stated envelope condition, the second-order remainder is controlled over the evaluated window, and we obtain

$$\widehat{\Delta}_{t,r} - \Delta_{t,r} = \left\{ r^{-1/2} \sum_{i=t-r+1}^{t} \partial_\vartheta s_i(\vartheta_0) \right\} h_n + R_{t,r}, \qquad \|R_{t,r}\| = O_p(\sqrt{r}\,\|h_n\|^2).$$

The mixing law of large numbers and central limit bound for the derivative process imply

$$r^{-1/2} \sum_{i=t-r+1}^{t} \partial_\vartheta s_i(\vartheta_0) = \sqrt{r}\, M_s + O_p(1), \qquad M_s = \mathbb{E}\{\partial_\vartheta s_i(\vartheta_0)\}.$$

We now distinguish the two rate conditions. Since $h_n = O_p(n_0^{-1/2})$, condition (a) makes the mean term $O_p\{(r/n_0)^{1/2}\} = o_p(1)$ and the remainder $O_p(\sqrt{r}/n_0) = o_p(1)$. Under condition (b), $M_s = 0$, so the linear term is $O_p(n_0^{-1/2})$, while $r/n_0^2 \to 0$ makes the remainder vanish. Thus $\widehat{\Delta}_{t,r} - \Delta_{t,r} = o_p(1)$ in either case.

For the information estimator, the information-envelope condition, consistency of $\widehat{\vartheta}$, and the mixing law of large numbers give $\widehat{I} - I = o_p(1)$. We conclude by continuous mapping and Slutsky's theorem. □

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